

AE-621A: Turbulence

Assignment 2

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Problem Statement

Using the provided hot-wire velocity data, compute the auto-correlation function $r(\tau)$ and estimate the integral time scale.

1 Theory

The instantaneous velocity in a turbulent flow is decomposed as

$$u(t) = \bar{U} + u'(t), \quad (1)$$

where $\bar{U}$ is the mean velocity and $u'(t)$ represents the turbulent fluctuation.

The normalized temporal auto-correlation function is defined as

$$r(\tau) = \frac{\langle u'(t) u'(t + \tau) \rangle}{\langle u'^2 \rangle}. \quad (2)$$

1.1 Physical Interpretation of the Auto-correlation Function

The auto-correlation function quantifies the statistical similarity between velocity fluctuations separated by a time lag τ , thereby measuring the temporal memory of turbulence.

1. **Zero lag** ($\tau = 0$): $r(0) = 1$, indicating perfect correlation.
2. **Small lag** ($0 < \tau \ll T_L$): $r(\tau)$ remains close to unity, implying strong coherence of turbulent eddies.
3. **Intermediate lag**: $r(\tau)$ decays gradually, reflecting progressive loss of memory in the flow.
4. **Zero crossing**: When $r(\tau) = 0$, fluctuations become statistically independent.
5. **Negative values**: $r(\tau) < 0$ indicates anti-correlation and possible oscillatory behaviour.

Thus, the decay of $r(\tau)$ from unity toward zero characterizes the lifetime and persistence of energy-containing turbulent structures.

2 Integral Time Scale

The integral time scale is defined as

$$T_L = \int_0^\infty r(\tau) d\tau. \quad (3)$$

For finite experimental data, the integration is performed numerically up to the first zero crossing of $r(\tau)$.

$$\boxed{T_L = 0.002152 \text{ s}} \quad (4)$$

The integral time scale represents the characteristic time over which turbulent velocity fluctuations remain correlated.

3 Results

The principal computed quantities are:

1. Mean Streamwise Velocity

$$\bar{U} = 3.2030 \text{ m/s}$$

Here, $\bar{U}$ represents the time-averaged velocity of the turbulent signal.

2. Sampling Time Step

$$\Delta t = 0.000250 \text{ s}$$

Here, Δt denotes the temporal resolution of the measurement.

3. Integral Length Scale

Using Taylor's frozen turbulence hypothesis,

$$L = \bar{U}T_L = (3.2030)(0.002152) \approx 6.89 \times 10^{-3} \text{ m}. \quad (5)$$

Here, L represents the characteristic size of the energy-containing turbulent eddies in the streamwise direction.

3.1 Auto-correlation Function (Short-Time Behaviour)

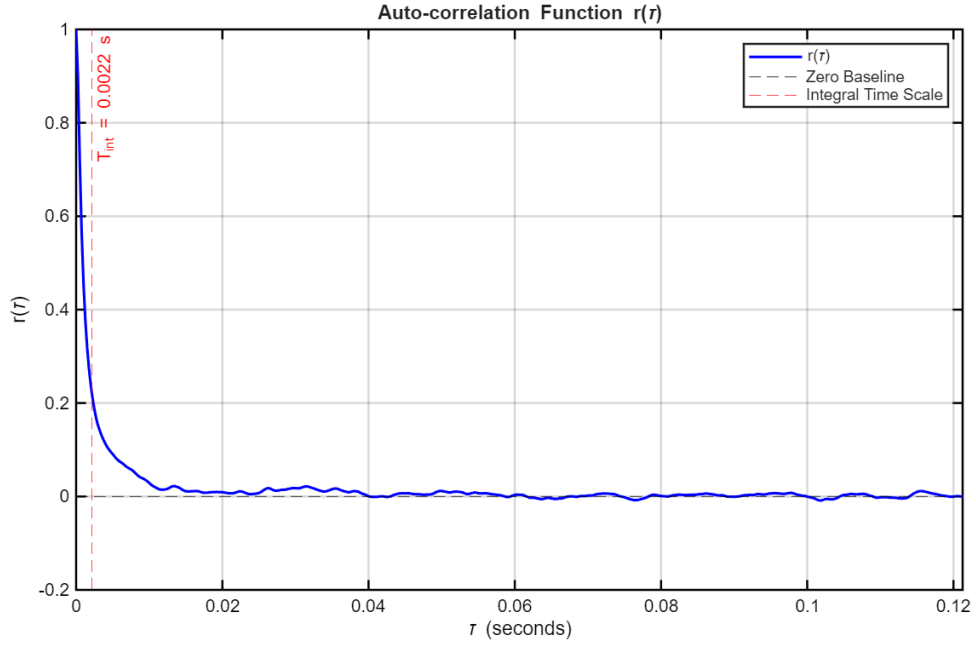


Figure 1: Auto-correlation function $r(\tau)$ for small time lags.

The auto-correlation decays rapidly from unity to zero within a few milliseconds, indicating fast temporal decorrelation of turbulent fluctuations. The area under this curve yields the integral time scale.

3.2 Long-Time Behaviour

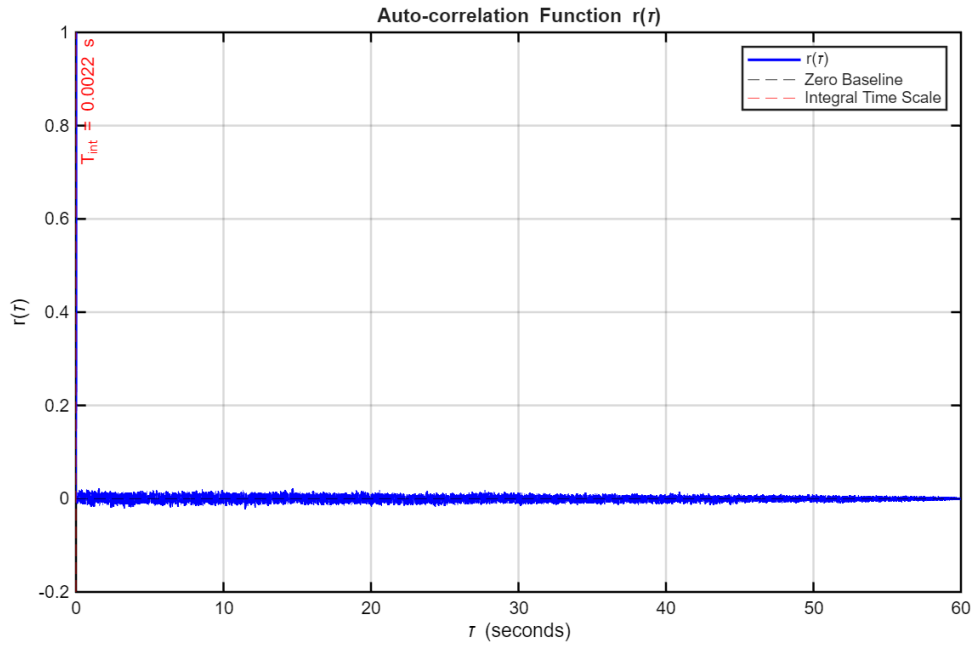


Figure 2: Auto-correlation over the full 60 s dataset.

Beyond small time lags, $r(\tau)$ fluctuates randomly around zero, confirming statistical independence of widely separated fluctuations.

4 Conclusion

The auto-correlation analysis reveals rapid temporal decorrelation of the turbulent velocity field. The estimated integral time scale of 0.002152 s corresponds to a characteristic length scale of approximately 6.89 mm, indicating short-lived energy-containing eddies.

The long-time behaviour further confirms statistical independence of fluctuations at large temporal separations.

Appendix A: MATLAB Implementation

The following MATLAB script was used to compute the auto-correlation function and estimate the integral time scale from the hot-wire data.

```

1 % Auto-correlation and Integral Time Scale Calculation
2
3 clear; clc; close all;
4
5 %% Load Data
6 data = load('hotwire1.dat');
7 time = data(:,1);
8 u_instant = data(:,2);
9
10 dt = time(2) - time(1);
11 N = length(u_instant);
12
13 %% Velocity Fluctuations
14 u_mean = mean(u_instant);
15 u_prime = u_instant - u_mean;
16
17 %% Auto-correlation
18 [autocorr_raw, lags] = xcorr(u_prime);
19
20 r_tau = autocorr_raw(lags>=0);
21 tau = lags(lags>=0)*dt;
22
23 r_tau = r_tau / r_tau(1); % Normalize
24
25 %% Integral Time Scale
26 zero_idx = find(r_tau<=0,1);
27 if isempty(zero_idx)
28     zero_idx = length(r_tau);
29 end
30
31 T_int = trapz(tau(1:zero_idx), r_tau(1:zero_idx));
32
33 %% Visualization
34 figure('Color','w');
35 plot(tau,r_tau,'LineWidth',1.5);
36 grid on;
37 xlabel('\tau (s)');
38 ylabel('r(\tau)');

```

```
39 yline(0,'k--');
40 xline(T_int,'r--');
41
42 %% Output
43 fprintf('Mean Velocity: %.4f m/s\n',u_mean);
44 fprintf('Integral Time Scale: %.6f s\n',T_int);
```